

Jennifer P. Lucero

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🌐 <https://astrojennz.github.io>

PhD student in Astrophysics with research experience in exoplanet detection and characterization with the radial velocity method and in high-resolution transmission spectroscopy.

🎓 Education

Instituto de Astrofísica e Ciências do Espaço, Universidade do Porto

PhD in Astrophysics

2023 – Ongoing

Porto, Portugal

- Current GPA: 18/20
- Thesis title: Detecting exoplanet atmospheres in the presence of stellar activity
- Supervisors: Olivier Demangeon & Nuno C. Santos

Pontificia Universidad Católica de Chile

BSc in Astronomy

2019 – 2022

Santiago, Chile

- GPA: 6.2/7 ([transcript](#))
- Thesis title: Optimized routines for exoplanet detection by calculating high-precision radial velocities
- Supervisor: James S. Jenkins

🔬 Research Interests

- Search and characterization of extrasolar planets
- Habitability of extrasolar planets
- Exoplanet atmospheres (modelling and observational)
- Stellar activity

📁 Research Experience

Project: Exploring the diversity of the most extreme exoplanetary systems

Research Assistant

2022 – 2023

Universidad Diego Portales

- Supervisor: James S. Jenkins
I developed an algorithm to measure high-precision radial velocities with the CHIRON spectrograph. Working on this algorithm would later become the main topic of my undergraduate thesis.

Project: Revision of the ESPRESSO calibration plan

Summer Research Intern

2023

European Southern Observatory, Chile

- Supervisors: Matias Jones & Elyar Sedaghati
For this project I reduced and analyzed ESPRESSO calibration data, in order to determine the adequate validity periods for each type of calibration, based on the stability requirements of the instrument.

Project: Study of short-period tidal inflation of exo-Neptunes

Summer Research Intern

2022

Pontificia Universidad Católica de Chile

- Supervisors: Cristobal Petrovich & Sarah Millholland
Using the Python package Kozaipy, I analyzed the dynamic inflation of exo-Neptunes by tidal effects as a function of time, obliquity, and eccentricity.

Project: VLT/MUSE IFU Observations of Interacting and Active Galaxies

Summer Research Intern

2020 – 2021

Pontificia Universidad Católica de Chile

- Supervisors: Ezequiel Treister & Giacomo Venturi
I studied the kinematics and ionization mechanisms of the gas composing the galaxy NGC 17, a system in the final fusion stage, using data cubes from the Multi Unit Spectroscopic Explorer (MUSE), on the VLT.

★ Observational Experience

La Silla Observatory

On site

2025

La Serena, Chile

- Observing astronomer with E. Cristo (NIRPS Consortium), supporting 9 nights of NIRPS & HARPS observations during GTO Run 13 with the 3.6m telescope.

Santa Martina Observatory

On site

2023

Santiago, Chile

- Worked as an astronomer assistant at the Observatory, where astronomy students do their scientific observations. I participated as MEADE30 and MEADE40 telescope operation.

Paranal Observatory

On site

2023

Antofagasta, Chile

- I accompanied Dr. Matias Jones for 10 nights at the Paranal Observatory, specifically at the VLT control building for UT3 telescope operation, as part of my Summer Internship at ESO.

📖 Teaching Experience

Pontificia Universidad Católica de Chile

Teaching & Outreach

2020-2023

Santiago, Chile

- Teaching Assistant in undergraduate Physics & Astronomy courses, including: General Physics II (FIS1504), Electromagnetic Theory (FIZ0321), Laboratory of Thermodynamics (FIS0152), Laboratory of Waves & Optics (FIZ312L), Astronomy (AST011), Astronomy (english version, AST0112), The Relativistic Universe of Einstein (AST102), Planets in the Universe (AST1529) and A Journey Through the Universe (AST101).
- Staff and Coordinator in the Welcome Commission for erasmus students.
- Tutor in MAIBuddy Program, helping erasmus students to get used to chilean culture and university life.
- Tutor in the Physics Faculty Mentoring Program of a group of first semester Physics students to help them integrate into university life.

📖 Conferences & Workshops

HORS3S

Poster: Impact of stellar spots on the high-resolution transmission spectra of a giant planet around a Sun-like star

2026

Granada, Spain

Exoplanets 6

Poster: Stellar activity as a limitation for high-resolution exoplanet atmosphere studies

2026

Porto, Portugal

Thirty Minute Talk (TMT) at ESO Vitacura

Talk: Effect of stellar spots on high-resolution transmission spectroscopy

2025

Santiago, Chile

Extremely Precise Radial Velocities (EPRV 6)

Poster: Effect of stellar non-occulted spots on high-resolution transmission spectroscopy

2025

Porto, Portugal

PLATO - ESP2025

Talk: Effect of stellar spots on the high-resolution transmission spectra of an Earth-like planet in the habitable zone of a Sun-like star

2025

Marseille, France

PoET Workshop # 2

Talk: Detecting exoplanet atmospheres in the presence of stellar activity

2024

Porto, Portugal

34º Encontro Nacional de Astronomia e Astrofísica

Talk: Retrieving the transmission spectrum of a simulated transiting planet

2024

Guimarães, Portugal

Online Seminar Series, Faculty of Physics at Pontificia Universidad Católica de Chile

Poster: VLT/MUSE IFU Observations of Interacting and Active Galaxies

2021

Santiago, Chile

ESPRESSO IOT Meeting, European Southern Observatory*Talk: Revision of the ESPRESSO calibration plan*

2023

Santiago, Chile

CATA Exoplanet Meeting, Cerro Calán Observatory*Talk: An algorithm to confirm exoplanets with CHIRON spectra*

2022

Santiago, Chile

Participation*Science & Commit, Universidad de Chile (2022), Sagan Exoplanet Summer Workshop, Caltech (2022), Writing and Communicating your Science, ESO Chile (2023)* **Manuscripts & Publications**

- 1 **P. Lucero, Jennifer**, et al., Stellar activity imprints on the high-resolution transmission spectrum of an Earth-like planet in the habitable zone: implications for ELT/ANDES, 2026 (*in prep*).
- 2 Dethier, W., et al. including **P. Lucero, J.**, Effect of stellar spot spectral lines in high-resolution transit spectroscopy, 2025, Astronomy & Astrophysics (*review in progress*).
- 3 **P. Lucero, Jennifer**, et al., [Effect of stellar spots on the high-resolution transmission spectra of a giant planet around a Sun-like star](#), 2026, Astronomy & Astrophysics.
- 4 Cristo, E., et al. including **P. Lucero, J.**, [SOAPv4: a new step towards the modelling of stellar signatures in exoplanet research](#), 2025, Astronomy & Astrophysics.
- 5 C. Santos, Nuno, et al. including **P. Lucero, J.**, [PoET: the Paranal solar ESPRESSO Telescope](#), 2024, ESO The Messenger.
- 6 Espinoza-Retamal Juan I., et al. including **P. Lucero, J.**, [HATS-38 b and WASP-139 b join a growing group of hot Neptunes on polar orbits](#), 2024, AAS Journals.
- 7 Mistry, P., et al., including **P. Lucero, J.** [VaTEST I: Validation of Sub-Saturn Exoplanet TOI-181b in Narrow Orbit from its Host Star](#), 2022, MNRAS.

 Fellowships & Grants**Bécalos Undergraduate Academic Excellence Scholarship**

2017-2018

ESO Summer Research Fellowship

2022

CAUP CIAAUP-22/2023-BI-D PhD Fellowship

2023-Ongoing

 Consortia & Awarded Telescope Time**Paranal solar ESPRESSO Telescope (PoET)**

2025

*Member of Science Team***Planets Beyond the Milky Way: Using Extragalactic Stars to Test the Limits of Planet Formation**

2023

Co-Investigator, Instruments: CHIRON, Time: 75h **Outreach****Comic Con Portugal**

2026

*I participated as an exhibitor at the Planetario do Porto stand to talk about spectroscopy of stars, the transits method to detect exoplanets, and astrobiology.***Universidade Junior**

2025

*I taught a class on exoplanets (in portuguese) for secondary school students at the Porto Planetarium.***Días Abertos, Faculdade de Ciências UP**

2025

I took part in the Open Days event, where I worked at the astronomy stand to talk about study opportunities at the University of Porto. I presented a demonstration of how we observe transit light curves to detect and characterize exoplanets. The interaction was in Portuguese and aimed at high school students.

Ignite IAstro Armamar

2024

I participated in the Ignite IAstro session, where I presented my research (in portuguese) in a fast and demanding format: 5 minutes with 20 auto-advancing slides of 15 seconds each. I talked about exoplanet atmospheres in a simple and accessible way, adapted for a general audience.

Golden Webinars, Pontificia Universidad Católica de Chile

2021

I participated as a panelist in a talk with Dr. André Maeder from the Geneva Observatory on "Exploration of Scale Invariant Effects in Cosmology in Relation with Dark Components"

Latinas NASA

2021

I participated as a panelist in a talk with three NASA women of Latino origin: Sandra Cauffman, Farisa Morales, and Adriana Ocampo, on the role of women in STEM. Media Appearance: NASA en Español, official YouTube channel: [Latinas@NASA](#)

 Organization**Exoplanets 6**

2024-Ongoing

Local Organizing Committee

Extreme Precise Radial Velocity (EPRV) 6

2024-2025

Local Organizing Committee

 Skills

Programming & Data Analysis: Python (Advanced), C, Octave, SQL

Fits file handling Instrumentation, Photometry, Data reduction (Python, ESORex/ESOReflex-ESPRESSO pipeline)

Astronomy Tools DS9, QFitsView, Exo-Striker, Batman, SOAPv4

Other Software Skills: $\LaTeX$, Pack Office(Word, Excel, PowerPoint), Adobe Photoshop & Lightroom.

 References

Nuno C. Santos Professor, Instituto de Astrofísica e Ciências do Espaço - nuno@astro.up.pt

Olivier Demangeon Researcher, Instituto de Astrofísica e Ciências do Espaço - olivier.demangeon@astro.up.pt

James S. Jenkins Professor, Institute of Astrophysical Studies, Universidad Diego Portales - james.jenkins@mail.udp.cl

Matias Jones Staff Astronomer, ESO Chile - mjones@eso.org

 Communication

Spanish Native

English C1

Portuguese C2